

Lab 2

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Problem 1

System 1 Analysis

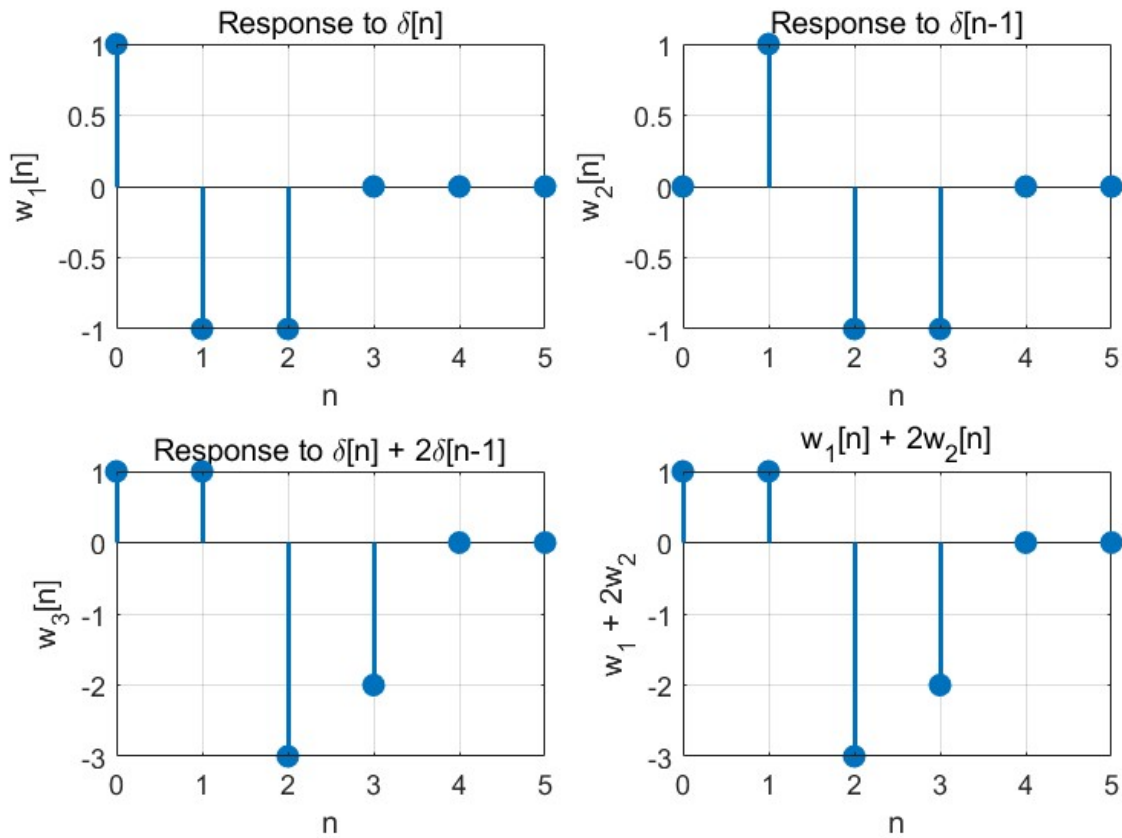


Figure 1: Figure for problem 1

From the MATLAB computation, we observe that

$$w_3[n] = w_1[n] + 2w_2[n], y_3[n] \neq y_1[n] + 2y_2[n], z_3[n] = z_1[n] + 2z_2[n].$$

1 Linearity

A system is linear if it satisfies the superposition principle:

$$T\{ax_1[n] + bx_2[n]\} = aT\{x_1[n]\} + bT\{x_2[n]\}$$

for any constants a, b and any inputs $x_1[n], x_2[n]$.

System 1

For any a, b and inputs $x_1[n], x_2[n]$:

$$T(ax_1 + bx_2) = (ax_1[n] + bx_2[n]) - (ax_1[n-1] + bx_2[n-1]) - (ax_1[n-2] + bx_2[n-2]) \quad (1)$$

$$= a[x_1[n] - x_1[n-1] - x_1[n-2]] + b[x_2[n] - x_2[n-1] - x_2[n-2]] \quad (2)$$

$$= aT(x_1) + bT(x_2) \quad (3)$$

Therefore, the system is linear.

System 2

Consider $x_1[n] = \delta[n]$ and $x_2[n] = \delta[n-1]$, with $a = 1, b = 2$.

Left-hand side (LHS):

$$T(\delta[n] + 2\delta[n-1]) = \cos(\delta[n] + 2\delta[n-1])$$

At $n = 0$:

$$y_3[0] = \cos(1 + 0) = \cos(1) \approx 0.5403$$

Right-hand side (RHS):

$$T(\delta[n]) + 2T(\delta[n-1]) = \cos(\delta[n]) + 2\cos(\delta[n-1])$$

At $n = 0$:

$$y_1[0] + 2y_2[0] = \cos(1) + 2\cos(0) = 0.5403 + 2 = 2.5403$$

Since $0.5403 \neq 2.5403$, the superposition principle fails. Therefore, System 2 is nonlinear.

System 3

For any constants a, b and inputs $x_1[n], x_2[n]$:

$$T(ax_1 + bx_2) = n \cdot (ax_1[n] + bx_2[n]) \quad (4)$$

$$= a(n \cdot x_1[n]) + b(n \cdot x_2[n]) \quad (5)$$

$$= aT(x_1) + bT(x_2) \quad (6)$$

This satisfies both homogeneity and additivity. Therefore, System 3 is linear.

2 Time-Invariance

A system is time-invariant if a time shift in the input results in an identical time shift in the output:

$$\text{If } y[n] = T\{x[n]\}, \text{ then } y[n - n_0] = T\{x[n - n_0]\} \text{ for any integer } n_0$$

Only System 3 is time-variant, because the coefficient n explicitly depends on time.

3 Summary

Table 1: Summary of Results

System	Input-Output Relation	Linearity	Time-Invariance
1	$w[n] = x[n] - x[n - 1] - x[n - 2]$	Linear	Time-Invariant
2	$y[n] = \cos(x[n])$	Non-Linear	Time-Invariant
3	$z[n] = n \cdot x[n]$	Linear	Time-Variant

Problem 2

1 Unit Step/Impulse Response

The unit step response and unit impulse response of the causal LTI system described by

$$\frac{dy(t)}{dt} + 3y(t) = x(t)$$

are:

$$s(t) = \frac{1}{3} (1 - e^{-3t}) u(t)$$

$$h(t) = e^{-3t} u(t)$$

The impulse response is the derivative of the step response:

$$h(t) = \frac{ds(t)}{dt} = e^{-3t} u(t)$$

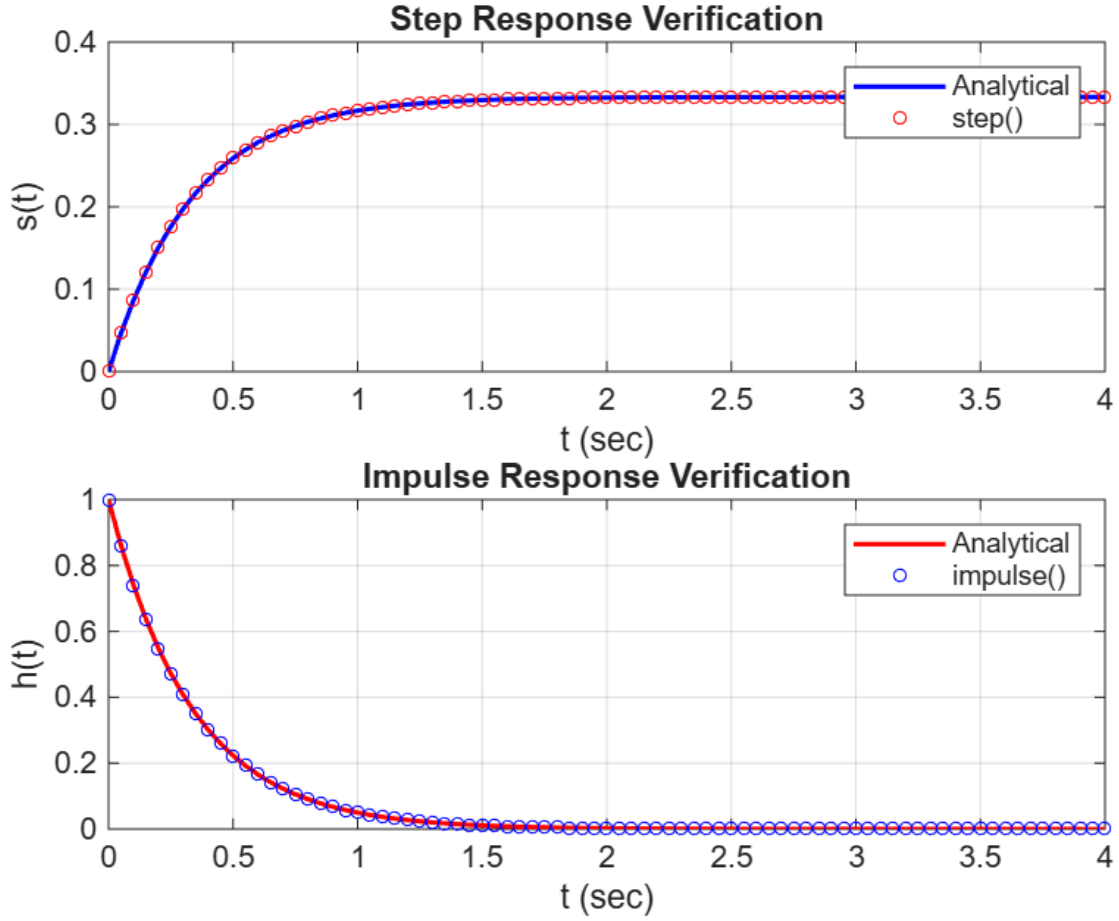


Figure 2: Unit step and unit impulse responses

2 Pulse Response $h^\Delta(t)$

The pulse $\delta^\Delta(t)$ is defined as:

$$\delta^\Delta(t) = \begin{cases} \frac{1}{\Delta}, & 0 \leq t < \Delta \\ 0, & \text{otherwise} \end{cases}$$

with unit area $\int_{-\infty}^{\infty} \delta^\Delta(t) dt = 1$.

We can derive that as Δ decreases, $h^\Delta(t)$ approaches $h(t)$. In the limit:

$$\lim_{\Delta \rightarrow 0} h^\Delta(t) = h(t)$$

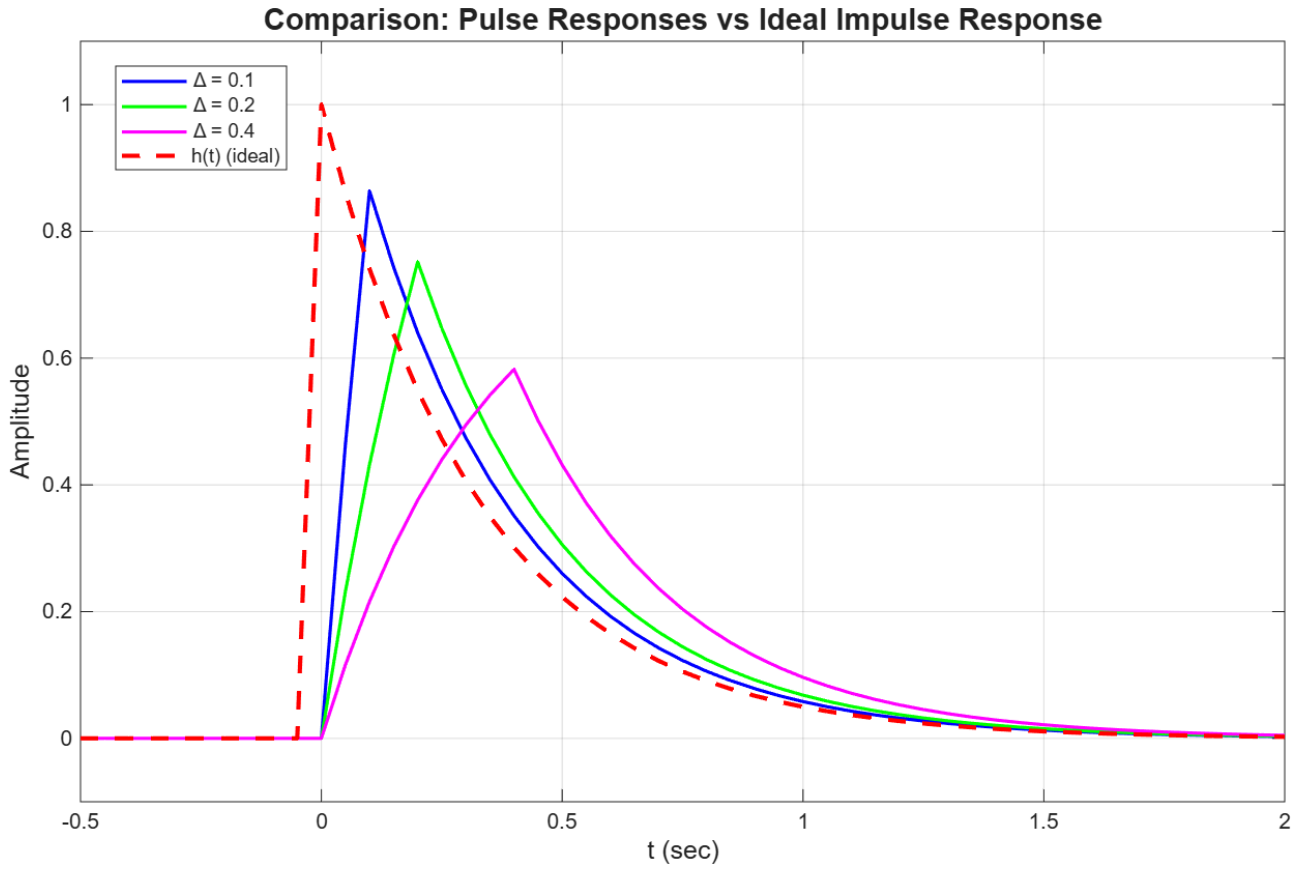


Figure 3: Pulse response

The pulse $D^\Delta(t)$ is defined as:

$$D^\Delta(t) = \begin{cases} 1, & 0 \leq t < \Delta \\ 0, & \text{otherwise} \end{cases}$$

with area $\int_{-\infty}^{\infty} D^\Delta(t) dt = \Delta$.

Since $\delta^\Delta(t) = \frac{1}{\Delta} D^\Delta(t)$ and the system is linear:

$$T\{D^\Delta(t)\} = \Delta \cdot T\{\delta^\Delta(t)\} \rightarrow \Delta \cdot h(t)$$

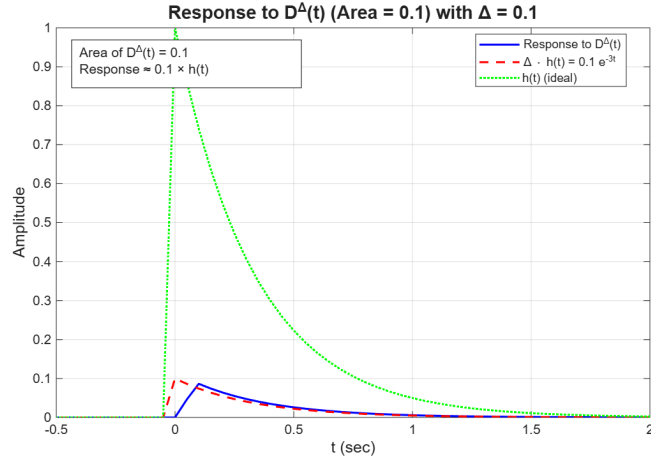


Figure 4: Pulse response when $\Delta = 0.1$

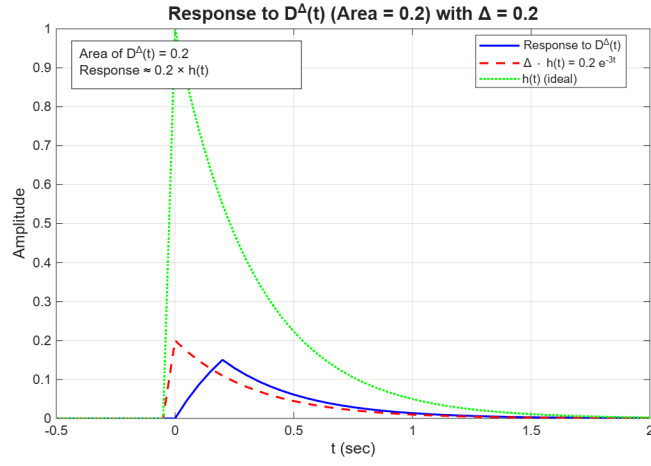


Figure 5: Pulse response when $\Delta = 0.2$

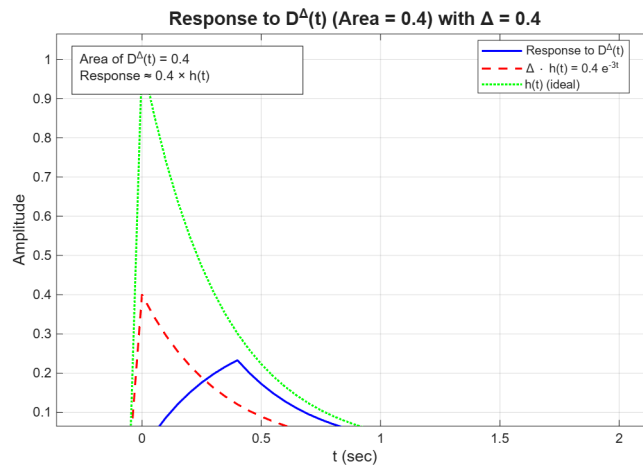


Figure 6: Pulse response when $\Delta = 0.4$

We can derive that unit area is necessary because:

- The Dirac delta $\delta(t)$ is defined to have unit area.

- To approximate $\delta(t)$, the pulse must also have unit area.
- If the pulse has area $A \neq 1$, the response will be $A \cdot h(t)$ instead of $h(t)$.

3 $d_a(t) = ae^{-at}u(t)$

The area of $d_a(t)$ is:

$$\int_{-\infty}^{\infty} d_a(t) dt = \int_0^{\infty} ae^{-at} dt = a \cdot \left[\frac{e^{-at}}{-a} \right]_0^{\infty} = a \cdot \left(0 - \left(-\frac{1}{a} \right) \right) = 1$$

Area of $d_a(t) = 1$ for all $a > 0$

Thus, $d_a(t)$ has unit area regardless of the value of a .

As $a \rightarrow \infty$:

- The function $d_a(t) = ae^{-at}u(t)$ becomes increasingly narrow
- The peak at $t = 0$ increases: $d_a(0) = a$
- The width decreases: most energy is in $[0, 4/a]$
- The area remains constant at 1

Therefore:

$$\lim_{a \rightarrow \infty} d_a(t) = \delta(t)$$

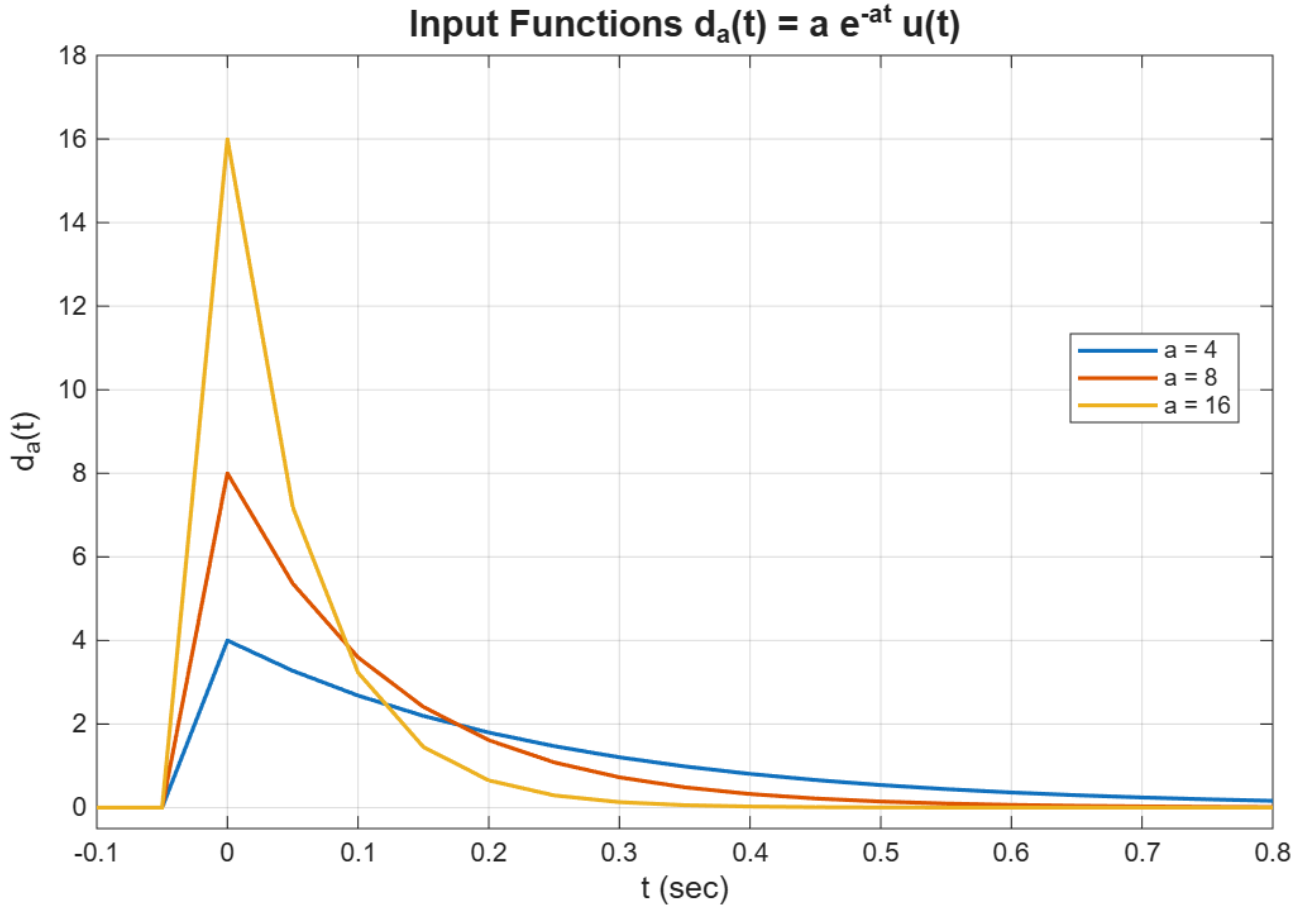


Figure 7: input function $d_a(t)$

The LTI system is described by:

$$\frac{dy(t)}{dt} + 3y(t) = x(t)$$

with impulse response $h(t) = e^{-3t}u(t)$.

Due to linearity and time-invariance, the response to $d_a(t)$ is the convolution:

$$y_a(t) = (d_a * h)(t) = \int_0^t a e^{-a\tau} \cdot e^{-3(t-\tau)} d\tau = e^{-3t} \int_0^t a e^{-(a-3)\tau} d\tau$$

For large a ($a \gg 3$), $d_a(t)$ approximates $\delta(t)$, so:

$$y_a(t) \approx h(t)$$

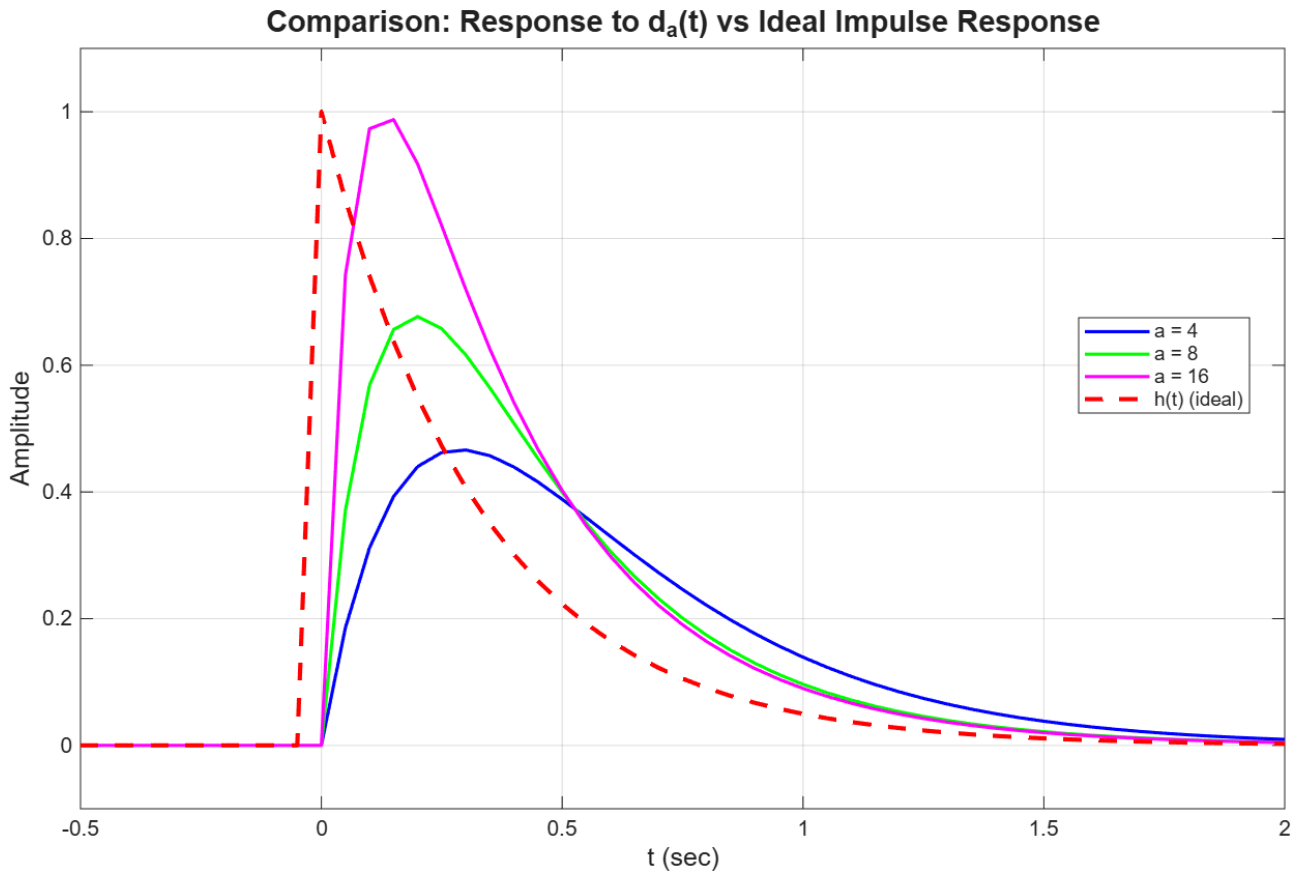


Figure 8: response to $d_a(t)$ for $a = 4, 8, 16$

As a increases:

- The input $d_a(t)$ becomes narrower and taller
- The response $y_a(t)$ approaches the ideal impulse response $h(t) = e^{-3t}u(t)$
- For $a = 16$, the response is practically identical to $h(t)$

For sufficiently large a ($a \gg 3$), the response to $d_a(t)$ is virtually indistinguishable from the true impulse response $h(t)$. This is because:

1. $d_a(t)$ has unit area for all a
2. As $a \rightarrow \infty$, $d_a(t) \rightarrow \delta(t)$
3. For an LTI system, $T\{\delta(t)\} = h(t)$

Problem 3

The echo system is described by:

$$y[n] = x[n] + \alpha x[n - N]$$

For an impulse input $x[n] = \delta[n]$, the output is the impulse response:

$$h_e[n] = \delta[n] + \alpha\delta[n - N]$$

With the given parameters $N = 1000$ and $\alpha = 0.5$:

$$h_e[n] = \delta[n] + 0.5\delta[n - 1000]$$

The impulse response has non-zero values only at:

- $n = 0$: $h_e[0] = 1$ (direct path)
- $n = 1000$: $h_e[1000] = 0.5$ (echo path)

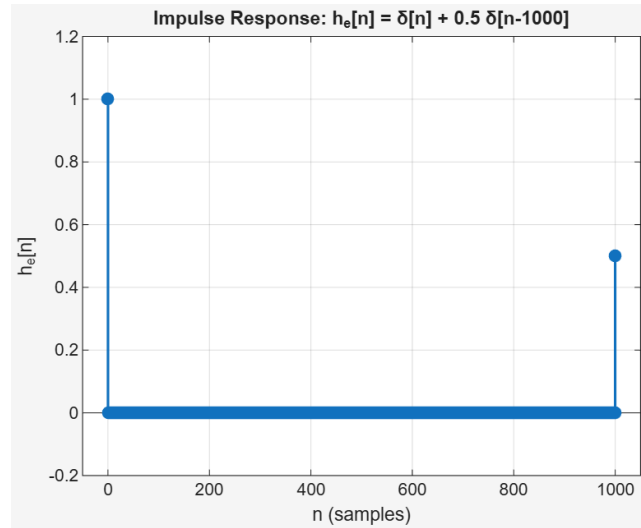


Figure 9: Echo system impulse response

The echo system is:

$$y[n] = x[n] + \alpha x[n - N] \quad (1)$$

$$z[n] + \alpha z[n - N] = y[n] \quad (2)$$

Substituting (1) into (2):

$$z[n] + \alpha z[n - N] = x[n] + \alpha x[n - N] \quad (7)$$

If we propose $z[n] = x[n]$ as a solution, then:

- Left-hand side: $x[n] + \alpha x[n - N]$
- Right-hand side: $x[n] + \alpha x[n - N]$

Since both sides are equal, $z[n] = x[n]$ is a valid solution.

Therefore, the inverse system described by $z[n] + \alpha z[n - N] = y[n]$ successfully removes the echo.

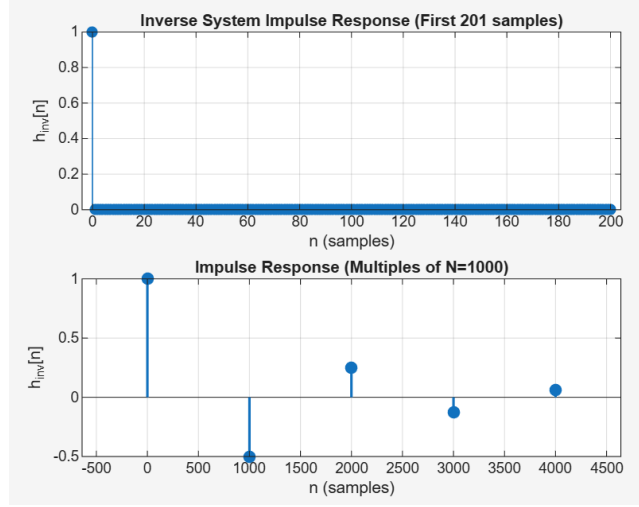


Figure 10: Inverse system

The inverse system is described by:

$$z[n] + \alpha z[n - N] = y[n]$$

This is equivalent to:

$$z[n] = y[n] - \alpha z[n - N]$$

The system is an IIR (Infinite Impulse Response) filter with coefficients:

- $\mathbf{a} = [1, 0, 0, \dots, 0, \alpha]$ (length $N + 1$)
- $\mathbf{b} = [1]$

The impulse response $h_{\text{inv}}[n]$ is obtained by setting $y[n] = \delta[n]$:

$$h_{\text{inv}}[0] = 1 \tag{8}$$

$$h_{\text{inv}}[N] = -\alpha = -0.5 \tag{9}$$

$$h_{\text{inv}}[2N] = \alpha^2 = 0.25 \tag{10}$$

$$h_{\text{inv}}[3N] = -\alpha^3 = -0.125 \tag{11}$$

$$\vdots \tag{12}$$

The general pattern is:

$$h_{\text{inv}}[k \cdot N] = (-\alpha)^k, \quad k = 0, 1, 2, \dots$$

All other samples are zero. This is an infinite-duration impulse response that decays exponentially as k increases (since $|\alpha| < 1$).

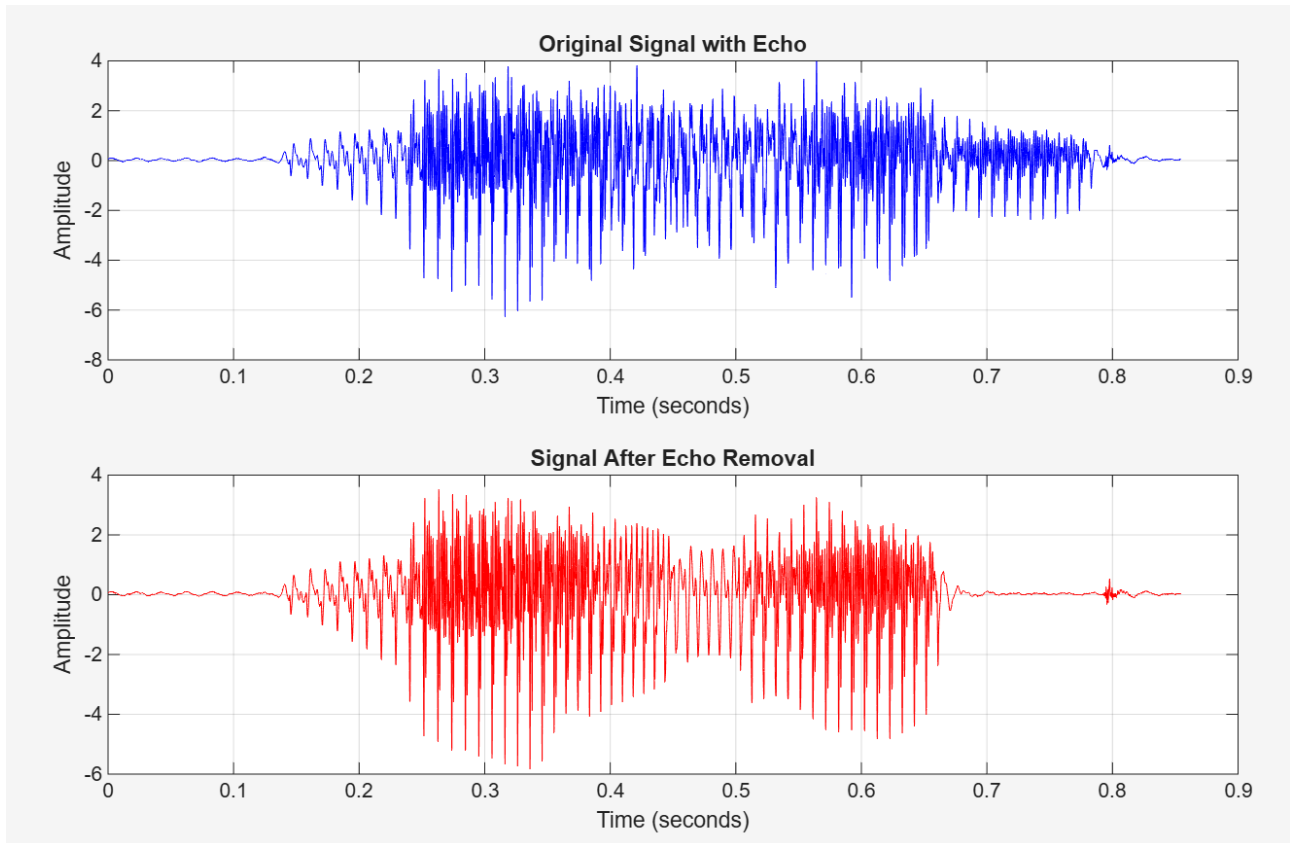


Figure 11: Inverse system impulse response

The echo removal system is implemented using the difference equation:

$$z[n] + \alpha z[n - N] = y[n]$$

This can be written as:

$$z[n] = y[n] - \alpha z[n - N]$$

Implementation:

- Input: $y[n]$ (speech signal with echo)
- Output: $z[n]$ (speech signal with echo removed)
- Parameters: $N = 1000$, $\alpha = 0.5$

Results:

- The original signal $y[n]$ contains the phrase "line up" with a noticeable echo.
- After processing through the inverse system, the echo is effectively removed.

- The output $z[n]$ sounds like the original uncorrupted speech.
- The echo cancellation is successful because the inverse system satisfies $z[n] = x[n]$.

Conclusion

The echo cancellation system is an inverse filtering approach that recovers the original speech signal from its echo-corrupted version. The key insight is that the echo system is a simple FIR filter, and its inverse is an IIR filter that can be implemented recursively. With proper choice of parameters ($N = 1000$, $\alpha = 0.5$), the inverse system effectively removes the echo while preserving the original speech quality.